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EXTENDED ABSTRACT

Spatially-contiguous reconstruction of water temperature and discharge in Switzerland using deep learning

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1. Introduction and objectives

Water temperature and discharge are critical variables for Switzerland's ecosystem health [Caissie, 2006] and energy production, particularly for nuclear cooling and hydroelectric production. These variables are physically coupled; low-flow conditions often exacerbate water temperatures [Booker and Whitehead, 2022], creating "compound events" that can threaten biodiversity or constrain energy supply.

Pressures on the hydrological system are intensified by anthropogenic climate change, e.g. through rising water temperature and decreasing summer discharge [Soto, 2018, Du et al., 2019, Michel et al., 2022, Tassone et al., 2023].

Switzerland's monitoring network of discharge gauges and water temperature sensors provides valuable insights into past and present conditions, but gaps in spatial coverage and record length still limit robust assessment of country-scale trends. To address this challenge, We develop a joint reconstruction of daily catchment-level water temperature and discharge from 1962 to 2023 using a Long Short-Term Memory (LSTM) network.

2. Methods and data

The LSTM is trained on 226 catchments to predict daily water temperature and discharge and requires precipitation and air temperature as meteorological inputs, besides static land properties. Meteorological features were derived from available gridded products developed by MeteoSwiss [MeteoSwiss, b, MeteoSwiss, a] and land properties from the MERIT [Yamazaki et al., 2017] and CORINE [European Environment Agency, 2019] datasets. To account for stability across seeds, we trained our network with ten different seeds. We assess the potential of this data-driven reconstruction through an exhaustive spatio-temporal cross-validation and evaluation at different temporal scales, ensuring that model fidelity is evaluated at times and locations that were not seen during training. In addition, we explore the potential of the multi-output architecture to decode physical coupling of water temperature and discharge and thereby improve representation of compound events.

3. Preliminary results

Our network achieves a median Nash-Sutcliffe efficiency of 0.91 for water temperature and 0.75 for discharge. The computational efficiency of our model enables an extended reconstruction with spatially contiguous predictions at 22000 locations along the Swiss river network. This first joint reconstruction of water temperature

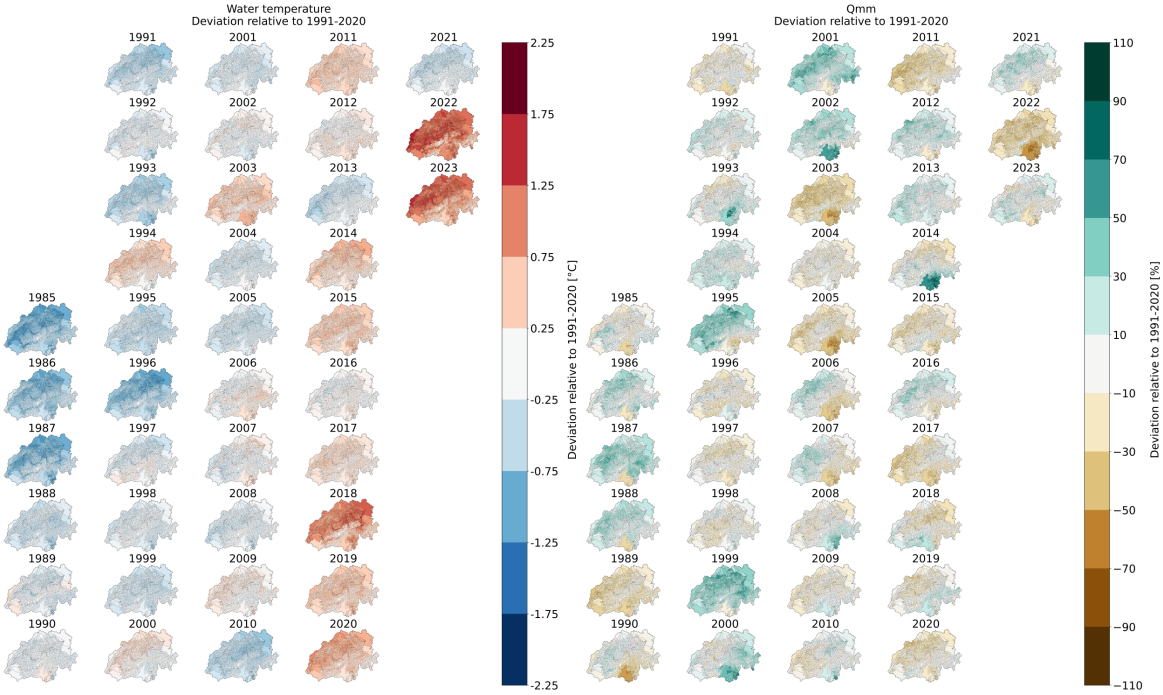

Figure 1. *Deviation from the 1991-2020 mean. Left: mean yearly water temperature deviation in °C. Right: accumulated yearly runoff deviation in %.*

and discharge for Switzerland opens avenues for process understanding and assessments of trends across Switzerland in the context of climate change. The results are promising and highlight potential for refining the model and expanding its applications, such as coupling with climate projections.

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